Micro Foundations of Financing Frictions: A Taxonomy

Class Note

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Abstract

This note provides a self-contained taxonomy of the main micro foundations of financing frictions used in macro-finance, development economics, and corporate finance. The core of the note is restricted to mechanisms that directly generate a wedge between the marginal revenue product of capital and the frictionless Walrasian cost of funds, or a borrowing limit that implies such a wedge when it binds. Two closely related mechanisms (incomplete-contract liquidation and liquidity/rollover risk) are treated as adjacent modules because their primitive distortion is ex post continuation or insurance rather than the initial marginal cost of capital.

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Frictionless Benchmark

Why a taxonomy of financing frictions? The marginal revenue product of capital varies widely across firms within narrowly defined industries, and that dispersion accounts for large aggregate TFP gaps across countries (Hsieh and Klenow, 2009). If capital flowed to its highest-value use, these gaps would close, so the dispersion is evidence that something stops it, and financing frictions are the leading explanation. This note asks which friction opens such a wedge and which policy closes it, and shows that the main candidates all reduce to one object, an MRPK wedge λ_i, arising from six distinct primitives, each with its own policy lever.

Purpose. Establish the departure point that every core friction distorts.

Setup.

- Technology: $y_i = z_i k_i^\alpha$, with $\alpha \in (0, 1)$.
- Capital market: competitive, all agents borrow and lend at the risk-free rate r. Throughout, r is the per-period user cost of capital, so one unit of capital costs r over the period and every lender break-even is written in these flow terms (not as a gross factor $(1+r)$).
- No asymmetric information, no enforcement problems, no market power, and complete contracts.

Firm problem. Normalize the output price to one, so $z_i k_i^\alpha$ is revenue. The entrepreneur with net worth a_i chooses capital k_i, financing the shortfall $k_i - a_i$ on the competitive capital market at rate r, to maximize profit:

$$\max_{k_i \geq 0} \; \pi_i = \underbrace{z_i k_i^\alpha}_{\text{revenue}} - \underbrace{r\,(k_i - a_i)}_{\text{cost of external finance}} - \underbrace{r\,a_i}_{\text{opportunity cost of net worth}} = z_i k_i^\alpha - r k_i.$$

Net worth a_i cancels: each unit of capital costs r whether it is borrowed or internal. This is the Modigliani and Miller (1958) logic in its sharpest form.

Optimal scale. The first-order condition equates the marginal revenue product of capital to its cost:

$$\underbrace{\alpha z_i k_i^{\alpha-1}}_{MRPK_i} = r \qquad \Longrightarrow \qquad k_i^* = \left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}}.$$

Result. Every firm reaches its frictionless scale, so the marginal revenue product of capital equals the common cost of funds, $\text{MRPK}_i = r$, for all i. Capital flows to its highest-value use, and net worth a_i is irrelevant for production decisions (Modigliani and Miller, 1958). Define the wedge $\lambda_i \equiv \text{MRPK}_i - r$; in this benchmark $\lambda_i = 0$ for every firm.

What must fail for $\lambda_i > 0$? At least one of the benchmark assumptions must fail: complete contracts, symmetric information, perfect enforcement, or competitive intermediation. The next section collects these departures into a single wedge.

Unifying Framework

Every core friction in this note reopens the wedge λ_i in the benchmark relation $\mathrm{MRPK}_i = r$, giving the same reduced-form object:

$$\mathrm{MRPK}_i = r + \lambda_i, \qquad \lambda_i \geq 0, \tag{1}$$

where r is the frictionless Walrasian cost of capital and λ_i is the shadow wedge that keeps firm i from choosing its frictionless scale. A firm with $\lambda_i > 0$ has a profitable marginal investment opportunity that is not financed at the frictionless cost of funds. The frictions differ in *why* $\lambda_i > 0$. The single organizing claim of this note is that these mechanisms share one reduced form but differ in the primitive that opens the wedge, and therefore in the policy lever that closes it; the sections that follow are arranged to make that contrast usable.

Some papers write the same object as a multiplicative wedge, $\mathrm{MRPK}_i = (1 + \tau_i)r$, with $\tau_i = \lambda_i/r$. This note uses the additive return wedge λ_i throughout.

Throughout, the output price is normalized to one, so $z_i k_i^{\alpha}$ is revenue and the marginal revenue product of capital is $\mathrm{MRPK}_i = \alpha z_i k_i^{\alpha-1}$. We read every optimality and wedge condition in per-period return terms: $\mathrm{MRPK}_i = r$ in the frictionless benchmark and $\mathrm{MRPK}_i = r + \lambda_i$ once a friction binds. In models with a linear capital technology, output is $R_i k_i$, the average and marginal return coincide at the gross per-period return R_i, and the same conditions read $R_i = r$ and $R_i = r + \lambda_i$.

Notation convention

How we read the cost of capital r and the return R_i throughout.

- r denotes the per-period user cost of capital, measured in the same units as output per unit of capital.
- It is the full frictionless cost of using one unit of capital for the period. It already summarizes any rental, depreciation, or financing cost needed for the model at hand.
- We do not distinguish between the risk-free interest rate, depreciation, and rental cost in this note. All of them are summarized by r.
- Lenders are risk-neutral and competitive, so contracts are priced by expected-value break-even. Thus r is the frictionless required return net of risk premia, and every λ_i in the note is a pure financing-friction wedge. If lenders are risk-averse, or if project payoffs carry priced aggregate risk, an observed gap between MRPK_i and r may reflect a risk premium rather than a distortion.
- R_i denotes output per unit of installed capital in linear technologies. In Cobb–Douglas technologies, the comparable object is the marginal revenue product, $\mathrm{MRPK}_i = \alpha z_i k_i^{\alpha-1}$.
- All wedge conditions are read in these same flow units: $\mathrm{MRPK}_i = r$ or $R_i = r$ in the frictionless benchmark, and $\mathrm{MRPK}_i = r + \lambda_i$ or $R_i = r + \lambda_i$ when a friction binds.

Conceptual key takeaway

What this note does not cover. This note explains where financing wedges come from. It does not solve the dynamic question of whether those wedges generate persistent aggregate misallocation. A **static wedge** is necessary for distorted investment at a point in time, but it is not sufficient for long-run misallocation.

In self-financing models, constrained firms may save, accumulate net worth, and eventually reach their efficient scale. **Persistent misallocation** typically requires forces that keep firms away from their self-financed steady state, such as idiosyncratic productivity shocks, entry and exit, borrowing constraints that bind after shocks, or uninsured entrepreneurial risk. The dynamic payoff of the models below is therefore not just that $\lambda_i > 0$, but whether the economy keeps recreating firms with high λ_i.

Primitive frictions versus reduced-form borrowing technologies

It is useful to separate:

- **Primitive frictions**: the deep information, enforcement, or market-structure problems that prevent firms from borrowing at the first-best rate and scale.
- **Reduced-form borrowing technologies**: the tractable constraints or contract forms, such as collateral ceilings, leverage caps, and external finance premia, that those primitive frictions generate in equilibrium.

A collateral constraint such as $b_i \leq \theta a_i$ or $b_i \leq \theta k_i$ is typically *not* a separate primitive. It is often the reduced-form consequence of deeper frictions such as moral hazard, limited enforcement, or adverse selection.

Conceptual key takeaway

The models below generate two reduced-form objects. Some produce a **quantity ceiling**: the firm wants to invest more at the going cost of funds but cannot. Others produce a **price wedge**: the firm can invest more only by paying a higher marginal cost of funds.

This has important implications for how to identify constrained firms in the data, and for how firms react to different shocks. We provide a clear distinction at the end of Friction 3.

Core versus adjacent mechanisms

The note keeps six mechanisms in the core taxonomy:

1. limited enforcement and collateral;
2. moral hazard and pledgeability;
3. costly state verification;

4. adverse selection and rationing;
5. intermediary balance-sheet constraints;
6. lender market power and hold-up.

Incomplete-contract liquidation and liquidity/rollover risk are important but slightly different. Their cleanest models produce inefficient continuation, liquidation, or insurance outcomes. They can be mapped into a marginal investment wedge only after adding an ex ante investment margin whose expected payoff is reduced by those ex post distortions. For that reason they are treated as adjacent modules rather than as two additional core input-cost frictions.

Organizational Map

#	Mechanism	What breaks	Side	Static/Dynamic	Anchor model
1	Limited enforcement / collateral	Commitment failure	Borrower	Dynamic	Kiyotaki–Moore (1997)
2	Moral hazard / pledgeability	Hidden action	Borrower	Either	Holmstrom–Tirole (1997)
3	Costly state verification	Hidden output	Borrower	Dynamic	Townsend (1979), BGG (1999)
4	Adverse selection / rationing	Hidden type or risk choice	Borrower	Static	Jaffee–Russell (1976), Stiglitz–Weiss (1981)
5	Intermediary constraints	Scarce bank capital	Lender	Dynamic	Holmstrom–Tirole (1997), Gertler–Kiyotaki (2010)
6	Lender market power / hold-up	Imperfect competition	Lender	Static	Rajan (1992), Sharpe (1990)
A	Incomplete-contract liquidation	Non-contractible states	Borrower/creditor	Dynamic	Aghion–Bolton (1992)
B	Liquidity / rollover risk	Missing future liquidity insurance	Market	Dynamic	Holmstrom–Tirole (1998)

Friction 1: Limited Enforcement and Collateral Constraints

The borrower cannot commit to repay. The lender's recourse is limited to seizing pledgeable assets.

(a) Definition

Borrowers can default or walk away from contracts, and courts can enforce repayment only up to the value of collateral. Lenders therefore restrict borrowing either to the pledgeable value of assets, or to a reduced-form multiple of borrower net worth when the collateral requirement is collapsed into a wealth ceiling.

(b) Seminal papers

- Kehoe and Levine (1993). "Debt-Constrained Asset Markets." *Review of Economic Studies* 60(4): 865–888.
- Hart and Moore (1994). "A Theory of Debt Based on the Inalienability of Human Capital." *Quarterly Journal of Economics* 109(4): 841–879.
- Kiyotaki and Moore (1997). "Credit Cycles." *Journal of Political Economy* 105(2): 211–248.

- Albuquerque and Hopenhayn (2004). "Optimal Lending Contracts and Firm Dynamics." *Review of Economic Studies* 71(2): 285–315.
- Banerjee and Newman (1993). "Occupational Choice and the Process of Development." *Journal of Political Economy* 101(2): 274–298.
- Evans and Jovanovic (1989). "An Estimated Model of Entrepreneurial Choice under Liquidity Constraints." *Journal of Political Economy* 97(4): 808–827.
- Rampini and Viswanathan (2010). "Collateral, Risk Management, and the Distribution of Debt Capacity." *Journal of Finance* 65(6): 2293–2322.
- Moll (2014). "Productivity Losses from Financial Frictions." *American Economic Review* 104(10): 3186–3221.
- Chaney, Sraer, and Thesmar (2012). "The Collateral Channel." *American Economic Review* 102(6): 2381–2409.

(c) Simplified model

Collateral borrowing limit. At date zero, the entrepreneur has net worth a_i, borrows b_i, and installs capital:

$$k_i = a_i + b_i.$$

We normalize the price of installed capital to one, so k_i denotes both the quantity of installed capital and its date-zero value. Let $\theta \in [0, 1]$ denote the fraction of installed capital value that the lender can recover after default. This implies that if the entrepreneur defaults, the lender can recover at most

$$\theta k_i.$$

The loan is settled in cash at the end of the period at the gross rate, principal plus interest:

$$(1 + r)b_i.$$

Capital is durable: it is retained (or resold) at end-of-period value k_i, which is why the per-period user cost is r and the benchmark profit is $z_i k_i^\alpha - r k_i$. The gross factor $(1 + r)$ enters only in the default decision below, where the borrower escapes the full cash repayment while keeping the physical capital.

If the entrepreneur defaults and walks away permanently, the lender seizes the recoverable fraction θk_i of the installed stock; the borrower keeps output and absconds with the unseizable capital $(1 - \theta)k_i$. Repayment is incentive compatible only if continuing is at least as good as walking away:

$$\underbrace{z_i k_i^\alpha + k_i - (1 + r)b_i}_{\text{payoff if the entrepreneur repays and continues}} \geq \underbrace{z_i k_i^\alpha + (1 - \theta)k_i}_{\text{payoff if the entrepreneur defaults and leaves}}.$$

Output cancels and the capital terms net to θk_i on the right, giving the limited-enforcement constraint:

$$(1 + r)b_i \leq \theta k_i.$$

Using the balance sheet identity,

$$k_i = a_i + b_i,$$

we get:

$$(1+r)(k_i - a_i) \leq \theta k_i,$$

so:

$$k_i \leq \bar{k}_i^{LE} \equiv \frac{(1+r)a_i}{(1+r)-\theta}.$$

Because $\theta \leq 1 < 1+r$, the denominator is always positive: the ceiling is well defined for every $\theta \in [0,1]$, with no knife-edge. Net worth enters because a_i reduces the amount that must be financed externally. A higher pledgeable fraction θ relaxes the constraint because more installed capital can be recovered after default. A higher interest rate r tightens it because the same amount borrowed carries a larger gross repayment. At $\theta = 0$ the firm can only self-finance, $\bar{k}_i^{LE} = a_i$; at $\theta = 1$ it reaches the maximal but still finite leverage $\bar{k}_i^{LE} = \frac{(1+r)a_i}{r}$. Full pledgeability of the capital stock does not restore the frictionless benchmark; it only maximizes feasible leverage.

Binding constraint and input wedge. In the frictionless benchmark,

$$k_i^* = \left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}}.$$

The collateral constraint binds whenever the pledgeable-capital ceiling is below the frictionless scale:

$$\frac{(1+r)a_i}{(1+r)-\theta} < \left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}},$$

or equivalently

$$a_i < \frac{(1+r)-\theta}{1+r}\left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}}.$$

When this condition holds, the firm chooses

$$k_i = \bar{k}_i^{LE} = \frac{(1+r)a_i}{(1+r)-\theta},$$

and the observed marginal product is

$$\text{MRPK}_i = \alpha z_i \left(\frac{(1+r)a_i}{(1+r)-\theta}\right)^{\alpha-1} = \alpha z_i \left(\frac{(1+r)-\theta}{(1+r)a_i}\right)^{1-\alpha} > r.$$

Thus the wedge is

$$\lambda_i^{LE} = \alpha z_i \left(\frac{(1+r)-\theta}{(1+r)a_i}\right)^{1-\alpha} - r.$$

The constraint is more likely to bind for firms with high productivity z_i, low net worth a_i, low pledgeability θ, or a high repayment rate r. Conditional on binding, the wedge is larger when z_i is higher, when a_i is lower, and when θ is lower. Higher pledgeability relaxes the constraint because more installed capital can support repayment. A higher repayment rate tightens the capital ceiling because the same amount borrowed requires a larger repayment.

(d) Which firms are constrained

Firms with low net worth relative to efficient scale and firms whose assets are hard to pledge: intangible-intensive, human-capital-intensive, services, software, and young firms. Asset-heavy firms with high liquidation value are less constrained for the same productivity and net worth.

(e) Policy lever

- Strengthening creditor rights and contract enforcement.
- Collateral registries, land titling, and movable-asset registries.
- Public guarantees that raise effective collateral value.
- Wealth transfers, start-up grants, or targeted credit lines when the main distortion is entry by high-ability, low-wealth entrepreneurs.

What does it take to be financially unconstrained

Holding productivity fixed, whether the firm reaches its frictionless scale depends on net worth a_i and the pledgeable fraction θ. The firm is unconstrained when $\bar{k}_i^{LE} \geq k_i^*$, that is when

$$a_i \geq \frac{(1+r)-\theta}{1+r}\left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}}.$$

For a given θ, higher net worth makes this easier; as $a_i \to \infty$ the firm self-finances any scale. For a given a_i, a higher pledgeable fraction θ raises the seizable stock and relaxes the ceiling.

Full pledgeability $\theta = 1$ does not by itself remove the constraint: it delivers the maximal but finite leverage $\bar{k}_i^{LE} = \frac{(1+r)a_i}{r}$, which still binds whenever k_i^* exceeds it. The deeper friction is that the lender can seize only installed capital, not the firm's future revenue; if revenue were fully seizable the borrower would never walk away and the wedge would vanish. Reaching the frictionless benchmark requires enough net worth or seizable returns, not merely $\theta = 1$ on the capital stock.

Notation: forms of the collateral constraint

A collateral constraint can be written in several forms that look equivalent but encode different timing and different economic objects. The single most important primitive separating them is **what the lender can seize when the borrower defaults**, because that decides whether net worth is invested *in* the project or posted *beside* it.

Productive asset seizable: Kiyotaki–Moore (1997), Hart–Moore (1994) (used in this section). The durable capital can itself be repossessed, so net worth is *invested*: the entrepreneur puts in a_i, borrows $b_i = k_i - a_i$, and the same installed capital that produces output is the collateral. The borrower owes the gross repayment $(1+r)b_i$ and can walk away permanently; the lender seizes a fraction θ of the stock, so incentive compatibility is $(1+r)b_i \leq \theta k_i$, giving $k_i \leq \frac{(1+r)a_i}{(1+r)-\theta}$. Net worth and collateral are the *same object*, and the gross factor appears because what walking away escapes is the full loan repayment, principal plus interest.

Productive asset not *seizable: Banerjee–Newman (1993).* If the project and its income cannot be reached, because the borrower flees with them, the project cannot serve as collateral at all. Net worth is then *posted beside* the project as a separate pledge, and the loan finances the *full* capital rather than only the shortfall $k_i - a_i$. With catch probability π and non-pecuniary punishment F, the limit is $L \leq w + \frac{\pi F}{r}$ (gross form $rL \leq rw + \pi F$), so the borrower can even borrow *beyond* his wealth, because punishment, not collateral alone, deters default. Here net worth and collateral are *separate objects*, and what is seizable is financial wealth, not the productive stock.

Intratemporal rental and diversion (Moll 2014, Buera–Kaboski–Shin 2011). The entrepreneur rents capital within the period and can divert a fraction before producing; the comparison is stock-on-stock, $k_i \leq \lambda a_i$ (equivalently $k_i \leq a_i/(1-\theta)$ with $\lambda = 1/(1-\theta)$), and no $(1+r)$ appears because no loan accrues interest across a period. The interest-free look is an artifact of that timing, not a claim that repayment carries no interest.

Reduced-form wealth ceiling. $b_i \leq \phi a_i$ implies $k_i \leq (1+\phi)a_i$; this names no seizable asset, it just makes own wealth scale feasible investment. Some papers (Itskhoki–Moll) posit the leverage multiple $k_i \leq \Lambda a_i$, $\Lambda \geq 1$, directly as a primitive rather than deriving it from a pledgeable fraction.

Pledgeable-income constraint. $\rho k_i \geq r(k_i - a_i)$ (Friction 2) says incentive-compatible cash flow must cover outside finance. In export-finance applications, $\xi_x R_i^x(k_i^x) \leq \Phi_x a_i$ limits working-capital financing per dollar of net worth.

All generate a binding scale ceiling and an MRPK wedge but point to different policy levers: collateral reform and creditor rights raise θ, financial development raises the leverage multiple λ (or Φ_x), stronger enforcement institutions raise the punishment term πF, and wealth transfers raise a_i.

Conceptual key takeaway

Friction 1, which we just saw, and Friction 2, which follows, are very similar in spirit and essentially deliver the same conclusion: the lender **rations the quantities of credit** because the borrower cannot credibly guarantee that if he defaults, the lender will recover the money lent.

In friction 1 (collateral constraint), the entrepreneur cannot commit to not "run away" with some of his capital if he defaults, and hence the lender can only recover a *fraction* of the productive capital invested.

Friction 2 (moral hazard) essentially "microfounds" *why* the entrepreneur would want to default in the first place and *why the lender will ask the entrepreneur to post some collateral in the first place.* The lender cannot perfectly monitor what the entrepreneur does once he lent him the money, and as a result the entrepreneur always has an incentive to limit his effort, which will lower the cash-flows he can produce.

This is why the entrepreneur cannot fully pledge his future cash-flows. There is always the risk that when the lender is not monitoring, the entrepreneur will shirk and lower his effort.

Key difference along two dimensions. First, what is not fully pledgeable differs:

- In friction 1, the limited "pledgeability" is about the value of the installed capital if the entrepreneur walks away.
- In friction 2, the limited "pledgeability" is about the effort the entrepreneur will make and hence about his future revenues.

Second, the deviation happens at a different stage:

- In friction 1, the deviation is at the repayment stage, after output is realized: the entrepreneur is tempted to default and walk away.
- In friction 2, the deviation happens before output is realized: even if final revenue were fully enforceable, so that the entrepreneur has no incentive to default, he may still exert low effort, because the private benefit from shirking is unobservable and therefore non-extractable.

Papers sometimes distinguish (1) and (2) by using "**limited enforcement**" for friction 1 and "**limited pledgeability**" for friction 2, but in practice, the two are often used interchangeably.

Conceptually, friction 2 "bites" before friction 1: If the entrepreneur was able to fully pledge his future cash-flows, the lender will not ask for any collateral (this is the frictionless benchmark, all $NPV > 0$ projects are financed). And it is *because* friction 2 exists that the question of the value of collateral in case of default is important.

Friction 2: Moral Hazard and Limited Pledgeability

The borrowing limit derived below can look like a reduced-form collateral or leverage cap, but here it comes from hidden action.

(a) Definition

Entrepreneurs can take hidden actions that reduce the probability of success while giving them private benefits. To induce good behavior, lenders must leave enough upside rents with the entrepreneur. This limits the income that can be pledged to outside financiers.

(b) Seminal papers

- Jensen and Meckling (1976). "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure." *Journal of Financial Economics* 3(4): 305–360.
- Holmstrom and Tirole (1997). "Financial Intermediation, Loanable Funds, and the Real Sector." *Quarterly Journal of Economics* 112(3): 663–691.
- Tirole (2006). *The Theory of Corporate Finance.* Princeton University Press, Chapters 3–4.

(c) Simplified model

Project. Investment k_i yields $R^g k_i$ in the good state and 0 in the bad state. High effort gives success probability p_H; shirking gives $p_L < p_H$ and private benefit Bk_i. Let $\Delta p = p_H - p_L$.

Maintained case. We focus on the classic moral-hazard case in which high effort is both socially efficient and delivers the highest pledgeable income to lenders. High effort is socially efficient if the extra success probability is worth the private benefit cost:

$$\Delta p \, R^g \geq B.$$

High effort also maximizes pledgeable income if the lender can pledge more expected income under incentive-compatible high effort than under low effort:

$$p_H \left(R^g - \frac{B}{\Delta p} \right) \geq p_L R^g.$$

Equivalently,

$$\Delta p \, R^g \geq p_H \frac{B}{\Delta p}.$$

Under these maintained inequalities, the relevant contract is the one that induces high effort. The lender therefore leaves the entrepreneur just enough upside to satisfy incentive compatibility and pledges the remaining expected income.

Incentive compatibility. If the entrepreneur receives R_E in the good state, high effort requires:

$$\underbrace{p_H R_E}_{\text{Payoff if high effort}} \geq \underbrace{p_L R_E + Bk_i}_{\text{Payoff if low effort}} \quad \Longrightarrow \quad R_E \geq \frac{Bk_i}{\Delta p}.$$

Pledgeable income. To maximize pledgeable income while inducing high effort, the lender leaves the entrepreneur exactly the minimum good-state payoff required by incentive compatibility:

$$R_E = \frac{Bk_i}{\Delta p}.$$

Good-state output is $R^g k_i$, so the maximum good-state payment that can be pledged to lenders is:

$$R^g k_i - R_E = R^g k_i - \frac{Bk_i}{\Delta p} = \left(R^g - \frac{B}{\Delta p}\right) k_i.$$

Under high effort, success occurs with probability p_H. Therefore maximum expected pledgeable income is:

$$p_H \left(R^g - \frac{B}{\Delta p}\right) k_i.$$

Dividing by investment k_i, pledgeable income per unit of capital is:

$$\rho_{MH} \equiv p_H \left(R^g - \frac{B}{\Delta p}\right).$$

Borrowing ceiling. Under the maintained high-effort contract, the maximum expected income pledgeable to lenders is $\rho_{MH} k_i$. Competitive lenders are willing to provide outside finance $k_i - a_i$ only if pledgeable income covers the required repayment:

$$\rho_{MH} k_i \geq r(k_i - a_i).$$

The interesting moral-hazard case is $\rho_{MH} < r$. Then each unit of capital generates less pledgeable income than the per-unit cost of funds. Entrepreneur net worth matters because it covers the unpledgeable gap. Rearranging gives:

$$(r - \rho_{MH}) k_i \leq r a_i,$$

so investment is bounded by:

$$k_i \leq \bar{k}_i^{MH} \equiv \frac{r a_i}{r - \rho_{MH}}. \tag{2}$$

Outside finance $k_i - a_i$ carries the per-period user cost r defined in the benchmark, so the break-even uses $r(k_i - a_i)$ rather than a gross factor $(1 + r)(k_i - a_i)$.

Binding constraint and input wedge. To translate the pledgeability ceiling into a production wedge, embed it in the concave technology from the benchmark. The frictionless scale is

$$k_i^* = \left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}}.$$

The moral-hazard constraint binds whenever the pledgeable-income ceiling is below the frictionless scale:

$$\frac{ra_i}{r-\rho_{MH}} < \left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}},$$

or equivalently

$$a_i < \frac{r-\rho_{MH}}{r}\left(\frac{\alpha z_i}{r}\right)^{\frac{1}{1-\alpha}}.$$

When this condition holds, the firm chooses

$$k_i = \bar{k}_i^{MH} = \frac{ra_i}{r-\rho_{MH}},$$

and the observed marginal product is

$$\text{MRPK}_i = \alpha z_i \left(\frac{ra_i}{r-\rho_{MH}}\right)^{\alpha-1} = \alpha z_i \left(\frac{r-\rho_{MH}}{ra_i}\right)^{1-\alpha} > r.$$

Thus the wedge is

$$\lambda_i^{MH} = \alpha z_i \left(\frac{r-\rho_{MH}}{ra_i}\right)^{1-\alpha} - r.$$

The constraint is more likely to bind for firms with high productivity z_i, low net worth a_i, or low pledgeable income ρ_{MH}. Conditional on binding, the wedge is larger when z_i is higher, when a_i is lower, and when ρ_{MH} is lower. Since

$$\rho_{MH} = p_H\left(R^g - \frac{B}{\Delta p}\right),$$

the constraint tightens when private benefits B are larger and relaxes when project returns R^g, monitoring quality, or incentive provision raise pledgeable income. A larger effort spread Δp also relaxes the constraint because it lowers the rent needed to induce high effort.

(d) Which firms are constrained

Firms with opaque cash flows, weak governance, or severe agency problems (high $B/\Delta p$) are constrained. The constraint binds hardest for firms with low net worth relative to efficient scale: young firms, R&D-intensive firms, intangible-capital firms, and firms in weak governance environments.

(e) Policy lever

- Monitoring by intermediaries, boards, or outside investors.
- Corporate governance reform and disclosure requirements that reduce private benefits B.
- Incentive compensation or contract design that increases the entrepreneur's cost of shirking.
- Pure collateral reform is not the direct lever unless it also changes pledgeability through incentives.

(f) What distinguishes this friction

The key parameter is $B/\Delta p$, not the liquidation value of assets. Moral hazard can produce a borrowing limit that resembles a collateral constraint, but the relevant policy comparative static is different: governance and monitoring relax this constraint; land titling alone need not.

Friction 3: Costly State Verification and the External Finance Premium

(a) Definition

The lender cannot freely observe the borrower's realized output. Verification is costly, so the optimal contract is standard debt with default and auditing. The deadweight verification cost generates an external finance premium that rises with leverage.

(b) Seminal papers

- Townsend (1979). "Optimal Contracts and Competitive Markets with Costly State Verification." *Journal of Economic Theory* 21(2): 265–293.
- Gale and Hellwig (1985). "Incentive-Compatible Debt Contracts: The One-Period Problem." *Review of Economic Studies* 52(4): 647–663.
- Bernanke and Gertler (1989). "Agency Costs, Net Worth, and Business Fluctuations." *American Economic Review* 79(1): 14–31.
- Bernanke, Gertler, and Gilchrist (1999). "The Financial Accelerator in a Quantitative Business Cycle Framework." In *Handbook of Macroeconomics*, Vol. 1.

(c) Simplified model

Orientation. This is the most involved model in the note. The algebra is longer than in the ceiling models because the contract must determine both a default threshold and an external finance premium. The payoff is a single object: an implied marginal wedge $\lambda_i^{CSV} = s(k_i/a_i; \gamma)$ that rises with leverage k_i/a_i and with the audit cost γ. A reader who wants only the taxonomy can take that and skip to (d); the derivation below shows where it comes from.

Destination. The derivation below earns the reduced form:

$$\lambda_i^{CSV} = s\left(\frac{k_i}{a_i}; \gamma\right), \qquad s_1(\cdot) > 0, \qquad s_2(\cdot) > 0.$$

Higher leverage k_i/a_i raises expected verification losses because the contract must pledge more output to lenders. A higher audit cost γ makes each default state more costly. The rest of this subsection derives this price wedge from the debt contract.

Project and timing. The entrepreneur has net worth a_i. To invest k_i, he must raise outside finance:

$$k_i - a_i.$$

The firm and lender choose the financing contract before uncertainty is realized. With Cobb–Douglas production, define average output per unit of capital as:

$$A_i(k_i) \equiv \frac{z_i k_i^\alpha}{k_i} = z_i k_i^{\alpha-1}.$$

Realized output is:

$$\omega_i A_i(k_i) k_i = \omega_i z_i k_i^\alpha.$$

The shock satisfies:

$$\omega_i \sim G, \qquad \omega_i \geq 0, \qquad \mathbb{E}[\omega_i] = \int_0^\infty \omega \, dG(\omega) = 1.$$

With this normalization, expected output before verification costs is:

$$\mathbb{E}[\omega_i A_i(k_i) k_i] = A_i(k_i) k_i = z_i k_i^\alpha.$$

After production, the firm observes ω_i freely. The lender observes ω_i only if it audits.

Average return versus marginal product. The CSV debt contract is naturally written in terms of average output per unit of capital, $A_i(k_i)$, because repayment is a claim on total realized output. The investment wedge in this note is still a marginal-product wedge. The marginal revenue product is:

$$\text{MRPK}_i(k_i) = \alpha z_i k_i^{\alpha-1} = \alpha A_i(k_i).$$

Conceptual key takeaway

The CSV contract splits total realized output, so it is written using average output per unit of capital, $A_i(k_i)$. The investment wedge in this note is still defined on the marginal product, MRPK_i. The factor between the two is α:

$$A_i(k_i) = z_i k_i^{\alpha-1}, \qquad \text{MRPK}_i(k_i) = \alpha z_i k_i^{\alpha-1} = \alpha A_i(k_i).$$

Debt contract. The contract specifies a default threshold $\bar{\omega}_i$, chosen before ω_i is realized. The promised repayment is:

$$\bar{\omega}_i A_i(k_i) k_i.$$

If $\omega_i \geq \bar{\omega}_i$, the firm repays without audit. If $\omega_i < \bar{\omega}_i$, the firm defaults and the lender audits.

Payoffs. In non-default states, the lender receives:

$$\bar{\omega}_i A_i(k_i) k_i,$$

and the entrepreneur keeps the residual:

$$(\omega_i - \bar{\omega}_i) A_i(k_i) k_i.$$

In default states, the lender audits and receives output net of verification costs:

$$(1-\gamma)\omega_i A_i(k_i)k_i,$$

while the entrepreneur receives zero.

Entrepreneur problem. The entrepreneur chooses the investment scale k_i and the default threshold $\bar{\omega}_i$ to maximize expected residual payoff:

$$\max_{k_i,\bar{\omega}_i} \int_{\bar{\omega}_i}^{\infty} (\omega - \bar{\omega}_i)A_i(k_i)k_i\, dG(\omega),$$

subject to lender break-even.

Lender break-even. Expected lender repayment is:

$$\underbrace{[1-G(\bar{\omega}_i)]\,\bar{\omega}_i A_i(k_i)k_i}_{\text{Repayment in non-default states}} + \underbrace{\int_0^{\bar{\omega}_i} (1-\gamma)\omega A_i(k_i)k_i\, dG(\omega)}_{\text{Recovery in default states}}.$$

Competitive lender break-even requires:

$$[1-G(\bar{\omega}_i)]\,\bar{\omega}_i A_i(k_i)k_i + \int_0^{\bar{\omega}_i} (1-\gamma)\omega A_i(k_i)k_i\, dG(\omega) = r(k_i - a_i).$$

Define the lender's expected repayment share as:

$$\Gamma(\bar{\omega}_i;\gamma) \equiv [1-G(\bar{\omega}_i)]\,\bar{\omega}_i + \int_0^{\bar{\omega}_i} (1-\gamma)\omega\, dG(\omega).$$

Then lender break-even is:

$$\Gamma(\bar{\omega}_i;\gamma)A_i(k_i)k_i = r(k_i - a_i),$$

or equivalently:

$$\Gamma(\bar{\omega}_i;\gamma)A_i(k_i) = r\left(1 - \frac{a_i}{k_i}\right).$$

Production return and required average return. The average-output object $A_i(k_i)$ in the debt contract is not a free choice variable. It must be generated by the production technology:

$$A_i^{prod}(k_i) = z_i k_i^{\alpha-1}.$$

Lender break-even implies a required average return:

$$A_i^{require}(k_i, \bar{\omega}_i) = \frac{r\left(1 - \frac{a_i}{k_i}\right)}{\Gamma(\bar{\omega}_i;\gamma)}.$$

Equilibrium requires:

$$A_i^{prod}(k_i) = A_i^{require}(k_i, \bar{\omega}_i).$$

Thus, for each investment scale k_i, lender break-even pins down the default threshold $\bar{\omega}_i$. The entrepreneur's choice can then be viewed as a choice of k_i, with the contract threshold adjusting to satisfy financing feasibility.

Recap of the logic. The entrepreneur chooses the investment scale k_i. Given k_i, the debt contract specifies the default threshold $\bar{\omega}_i$ that satisfies lender break-even. The production side pins down the total output available to split between the entrepreneur and the lender. This prevents the model from treating the average return as a free variable and rules out the artificial case in which investment is pushed to $k_i \to \infty$ for a fixed contract threshold.

Binding constraint. In this model, the binding constraint is the lender's break-even condition, not a simple ceiling $k_i \le \bar{k}_i$. Together with production, lender break-even determines the feasible pair $(k_i, \bar{\omega}_i)$. The note's wedge remains a marginal-product wedge:

$$\lambda_i^{CSV} = \mathrm{MRPK}_i(k_i) - r = \alpha z_i k_i^{\alpha-1} - r.$$

The two endogenous objects are k_i and $\bar{\omega}_i$. The threshold is chosen ex ante as part of the financing contract, not after the firm observes ω_i.

Comparative statics.

A firm with higher leverage k_i/a_i must pledge more expected output to the lender. This requires a higher default threshold $\bar{\omega}_i$. When $\gamma > 0$, a higher threshold raises expected verification costs, so outside finance becomes more expensive and the chosen scale is lower. The implied marginal wedge can be summarized as:

$$\lambda_i^{CSV} = s\left(\frac{k_i}{a_i}; \gamma\right), \qquad s_1(\cdot) > 0, \qquad s_2(\cdot) > 0.$$

The wedge increases with k_i/a_i because a more leveraged firm must pledge more expected output to lenders. It decreases with net worth a_i, holding k_i fixed, because the firm needs less outside finance. The wedge increases with the audit cost γ because default states destroy more output.

Optional check (skippable): the zero-friction case $\gamma = 0$. When verification is costless the contract only splits output and destroys nothing, so $\lambda_i^{CSV} = 0$; the algebra below verifies this. The lender's expected repayment share is:

$$\Gamma(\bar{\omega}_i; 0) = [1 - G(\bar{\omega}_i)]\,\bar{\omega}_i + \int_0^{\bar{\omega}_i} \omega\, dG(\omega).$$

The entrepreneur's expected residual share is:

$$\Pi_E(\bar{\omega}_i) = \int_{\bar{\omega}_i}^{\infty} (\omega - \bar{\omega}_i)\, dG(\omega).$$

Adding the two shares gives total expected output:

$$\Gamma(\bar{\omega}_i; 0) + \Pi_E(\bar{\omega}_i) = \int_0^{\infty} \omega\, dG(\omega) = 1.$$

Thus, when $\gamma = 0$, the contract only splits output between lender and entrepreneur. It does not destroy resources. Using lender break-even,

$$\Gamma(\bar{\omega}_i; 0) A_i(k_i) k_i = r(k_i - a_i),$$

the entrepreneur's expected payoff becomes:

$$\Pi_E(\bar{\omega}_i) A_i(k_i) k_i = z_i k_i^\alpha - r(k_i - a_i).$$

So the entrepreneur internalizes the full expected output from investment net of the competitive financing cost. The investment condition is frictionless:

$$\alpha z_i k_i^{\alpha-1} = r,$$

or equivalently:

$$\text{MRPK}_i = r.$$

Therefore:

$$\lambda_i^{CSV} = 0.$$

When $\gamma > 0$, default states destroy resources:

$$\Gamma(\bar{\omega}_i; \gamma) + \Pi_E(\bar{\omega}_i) = 1 - \gamma \int_0^{\bar{\omega}_i} \omega \, dG(\omega) < 1.$$

This expected verification loss is the resource cost behind the external finance premium.

(d) Which firms are constrained

Highly leveraged firms and firms whose output is hard to verify. This wedge is a continuous price distortion rather than a hard ceiling: externally financed firms pay a premium, and the premium is steepest for low-net-worth firms.

(e) Policy lever

- Accounting standards, disclosure, and standardized financial reporting.
- Auditing infrastructure and legal penalties for misreporting.
- Policies that raise net worth or retained earnings, shifting firms down the premium schedule.
- Collateral reform is indirect unless it reduces verification losses or leverage.

(f) What distinguishes this friction

Unlike in the moral hazard setting, the entrepreneur does not choose hidden effort; the problem is that realized output is costly to verify. Unlike collateral constraints, the distortion is usually a smooth external finance premium. The dynamic signature is the financial accelerator: net worth falls, the premium rises, investment falls, and net worth falls further.

Bridge: capital ceilings versus price wedges

A useful distinction is between the object that is productive and the object that relaxes financing. Productive capital k_i enters the production function. Entrepreneur net worth a_i is internal wealth, or equity, that reduces the amount of outside finance required to install k_i.

There are two different ways to model the friction. The first is a **capital ceiling**: the firm can invest only up to a maximum scale pinned down by collateral or net worth. The second is a **price wedge**: the firm can still choose any k_i, but external finance becomes more expensive as leverage rises.

The two capital-ceiling formulations are a collateral constraint on installed capital, $(1+r)b_i \leq \theta k_i$, and a reduced-form wealth ceiling, $b_i \leq \phi a_i$. The price-wedge formulation is a leverage premium, $\lambda_i = s(k_i/a_i)$.

If lenders can seize a fraction θ of installed capital and the borrower owes the gross repayment, the primitive collateral constraint is written on k_i:

$$\begin{aligned} &(1+r)b_i \leq \theta k_i, \qquad k_i = a_i + b_i, \\ &\implies \quad [(1+r)-\theta]k_i \leq (1+r)a_i \quad \implies \quad k_i \leq \frac{(1+r)a_i}{(1+r)-\theta}. \end{aligned} \tag{3}$$

In this formulation, net worth appears in the final upper bound because the entrepreneur must finance the share of capital that gross repayment cannot pledge against seizure. The lender's recourse is still installed capital, not net worth itself. The intratemporal-diversion variant replaces this with $k_i \leq a_i/(1-\theta)$; see the notation box in Friction 1.

By contrast, a reduced-form wealth constraint writes borrowing capacity directly as a multiple of net worth:

$$\begin{aligned} &b_i \leq \phi a_i, \qquad k_i = a_i + b_i, \\ &\implies \quad k_i \leq (1+\phi)a_i. \end{aligned} \tag{4}$$

This form does not specify which productive asset the lender can seize. It summarizes a financing technology in which own wealth scales feasible investment.

A third formulation keeps the quantity choice unconstrained but makes external finance more expensive when leverage is high:

$$\lambda_i = s\left(\frac{k_i}{a_i}\right), \qquad s'(\cdot) > 0. \tag{5}$$

This is a price wedge, not a capital ceiling. Net worth lowers the wedge by reducing leverage, rather than imposing a hard upper bound on k_i.

Capital ceiling equivalence. In a one-period model the two hard quantity constraints are mechanically equivalent: using the balance sheet identity $k_i = a_i + b_i$, the collateral ceiling (3) gives $k_i \leq \frac{(1+r)a_i}{(1+r)-\theta}$ and the wealth ceiling (4) gives $k_i \leq (1+\phi)a_i$, so both reduce to a common maximum feasible scale $\bar{k}(a_i)$ and the firm solves

$$k_i = \min\left\{k_i^*, \bar{k}(a_i)\right\}.$$

Price wedge. The leverage-premium formulation is different. It does not imply a capital ceiling. Instead, it keeps k_i as an unconstrained choice and changes the marginal cost of external finance:

$$\lambda_i = s\left(\frac{k_i}{a_i}\right), \qquad s'(\cdot) > 0.$$

Higher k_i/a_i raises the financing wedge, so the firm chooses a lower scale because capital is more expensive at the margin, not because it hits a hard borrowing limit.

Empirical implications

The wedge $\lambda_i = \text{MRPK}_i - r$ is observable in different places depending on its source.

- **Quantity rationing** (Frictions 1, 2, 4). The firm is rationed: it would borrow more at the going rate but cannot. The wedge is a shadow cost it never pays in the loan rate, so two firms at the same rate can face very different λ_i. It is identified only from the gap $\text{MRPK}_i - r > 0$, that is, from dispersion in the marginal revenue product of capital, not from the rate.
- **Price wedge** (Frictions 3, 5, 6). The firm actually pays $r + \lambda_i$, so the wedge appears directly in the observed lending rate or spread (firm-specific and dispersed across firms under costly state verification; common across affected borrowers under the intermediary and market-power frictions, identified by comparing exposed firms with unexposed ones).

Both regimes can generate the same $k \propto a$ reduced form; an empiricist tells them apart by whether the distortion sits in the price of credit or only in the quantity.

Conceptual key takeaway

- Under quantity rationing the firm sits at its borrowing limit, off the frictionless first-order condition ($\text{MRPK}_i > r$), and cannot expand even when investment opportunities improve. It is at a corner, not optimizing the marginal unit.
- Under a price wedge the firm is at an interior optimum on its demand curve, $\text{MRPK}_i = r + \lambda_i$ with equality, and substitutes smoothly as the wedge moves.

Both deliver the same underinvestment; they differ in how the firm responds to a change in opportunities or in the wedge.

Dynamic difference. The equivalence is less innocuous in dynamic models because a_i is typically predetermined at the start of the period, while k_i is a choice variable and collateral value may depend on future equilibrium objects. A wealth ceiling gives a simple state-dependent feasible set for the dynamic program: solve for $k_i(a_i, z_i)$ subject to an upper bound that is known when the firm chooses investment. A collateral constraint such as

$$b_{i,t} \leq \theta q_t k_{i,t} \qquad \text{or} \qquad b_{i,t} \leq \theta \mathbb{E}_t[q_{t+1} k_{i,t+1}]$$

requires solving for collateral values jointly with policy functions, asset prices, liquidation values, or continuation values. This extra structure is valuable when the model is about collateral-price feedback, creditor rights, or liquidation values; it is not always needed when the goal is simply to capture how low net worth limits scale.

The distinction matters for interpretation and comparative statics. In a pledgeable-capital model, reforms that raise the seizable value of installed assets, such as creditor rights or collateral registries, directly relax the constraint through θ. In a wealth-based reduced form, policies that raise net worth, retained earnings, or start-up wealth directly relax the constraint through a_i. In a leverage-premium model, the same increase in a_i reduces the marginal cost of external finance smoothly rather than moving a hard investment ceiling.

Transition: from borrower-side frictions to lender-side frictions

The first three frictions focus on why a borrower cannot pledge enough income or collateral. The next three frictions put more structure on the lender side. The bank may choose a lending rate, ration credit, face its own balance-sheet constraint, or exploit market power.

A useful reduced-form lender problem is:

$$\max_{\hat{r}} \left[m(\hat{r}) - r\right] L(\hat{r}),$$

where $\hat{r}$ is the promised loan rate, r is the bank's cost of funds, $m(\hat{r})$ is the expected repayment per unit lent, and $L(\hat{r})$ is loan volume.

Different lender-side frictions change different parts of this problem:

- **Adverse selection.** Raising $\hat{r}$ worsens borrower composition, so expected repayment $m(\hat{r})$ may be non-monotone. The lender may prefer to keep the rate at $\hat{r}^*$ and ration credit rather than raise the rate.
- **Intermediary balance-sheet constraints.** The bank may want to lend more at the prevailing spread, but total lending is limited by bank net worth or regulation.
- **Market power and relationship lending.** The lender chooses a markup because loan demand is downward sloping or because borrowers are locked into a relationship.

This transition helps clarify what changes in Friction 4. The adverse-selection model is not just a model where a borrower has a constraint. It is the first core model in the note where the lender's loan-rate choice changes the composition of borrowers. By contrast, in the CSV model the lender and borrower choose a contract to handle costly verification, but the contract does not change the ex ante borrower pool.

Friction 4: Adverse Selection and Credit Rationing

(a) Definition

Lenders cannot distinguish borrower types. Raising the promised repayment can worsen the borrower pool because safer borrowers drop out first. The lender's expected repayment can therefore be non-monotone in the loan rate, so credit may be rationed at a rate below the market-clearing rate. Stiglitz–Weiss also emphasizes a second channel, in which higher rates induce borrowers to choose riskier projects; the simplified model below isolates the borrower-composition channel.

(b) Seminal papers

- Akerlof (1970). "The Market for 'Lemons'." *Quarterly Journal of Economics* 84(3): 488–500.
- Jaffee and Russell (1976). "Imperfect Information, Uncertainty, and Credit Rationing." *Quarterly Journal of Economics* 90(4): 651–666.
- Stiglitz and Weiss (1981). "Credit Rationing in Markets with Imperfect Information." *American Economic Review* 71(3): 393–410.
- Bester (1985). "Screening vs. Rationing in Credit Markets with Imperfect Information." *American Economic Review* 75(4): 850–855.
- Myers and Majluf (1984). "Corporate Financing and Investment Decisions When Firms Have Information That Investors Do Not Have." *Journal of Financial Economics* 13(2): 187–221.
- de Meza and Webb (1987). "Too Much Investment: A Problem of Asymmetric Information." *Quarterly Journal of Economics* 102(2): 281–292.

(c) Simplified model

Lender problem. In the adverse-selection model, loan volume is fixed at available loan supply $\bar{L}$, so the lender chooses the promised repayment $\hat{r}$ to maximize expected profit per unit lent:

$$\max_{\hat{r}} \left[\Pi(\hat{r}) - r\right] \bar{L}.$$

Since $\bar{L}$ is fixed, the lender chooses $\hat{r}$ to maximize expected repayment $\Pi(\hat{r})$, subject to breaking even. The key point is that $\Pi(\hat{r})$ need not increase with $\hat{r}$, because a higher promised repayment worsens the borrower pool. If loan demand exceeds $\bar{L}$ at the maximizing rate $\hat{r}^*$, the lender rations credit rather than raising the rate. Rationed firms receive less capital than their frictionless desired scale, $k_i < k_i^*$.

Borrower types. There are safe and risky borrowers. Safe borrowers generate enough cash flow to repay with certainty up to a threshold. Risky borrowers generate high payoff with probability p and zero otherwise. The lender cannot observe type.

Lender return. At promised repayment $\hat{r}$ per unit lent, the lender receives $\hat{r}$ from safe borrowers and $p\hat{r}$ from risky borrowers. If $\mu(\hat{r})$ is the fraction of safe borrowers who apply, expected repayment is:

$$\Pi(\hat{r}) = \left[\mu(\hat{r}) + (1 - \mu(\hat{r}))p\right] \hat{r}.$$

Because safe borrowers exit as $\hat{r}$ rises, $\mu'(\hat{r}) < 0$. Therefore $\Pi(\hat{r})$ can decline when the rate rises.

Rationing equilibrium. Let $\hat{r}^*$ maximize the lender's expected return. If loan demand exceeds loan supply at $\hat{r}^*$, lenders do not raise the rate because doing so worsens expected repayment. They ration credit instead. The model pins down aggregate credit supply $\bar{L}$, not a unique firm-level allocation. If n observationally eligible firms are rationed symmetrically, a simple allocation rule is $\bar{k}_i = \bar{L}/n$; other rationing rules can generate different $\bar{k}_i$'s, but the key point is that each rationed borrower receives less than its desired frictionless scale.

Input wedge. A rationed good borrower receives $k_i < k_i^*$ or no loan at all. If it receives a binding allocation $\bar{k}_i < k_i^*$, then:

$$\lambda_i^{AS} = \alpha z_i \bar{k}_i^{\alpha-1} - r > 0. \tag{6}$$

As in the quantity-rationing column of the framework, λ_i^{AS} is a shadow cost, not a loan rate the firm pays.

(d) Which firms are constrained

In the pure Stiglitz–Weiss model, rationing can be random among observationally identical applicants. In richer models with partial screening, rationing falls disproportionately on opaque firms, young firms without track records, and firms lacking pledgeable signals. Bester (1985) shows that collateral can screen types, so collateral requirements may be a screening device rather than only an enforcement device.

(e) Policy lever

- Credit registries and information sharing.
- Subsidized screening or certification.
- Carefully designed public guarantees that do not worsen pool composition.
- Collateral or relationship lending as screening technologies.

(f) What distinguishes this friction

The wedge comes from hidden type or hidden risk choice, not hidden effort or limited enforcement. A pure injection of loanable funds may not eliminate rationing if it leaves borrower composition unchanged. The policy problem is to improve screening or risk allocation, not simply to subsidize rates.

Friction 5: Intermediary Balance-Sheet Constraints

(a) Definition

Even when borrowers are creditworthy, the intermediary's own net worth can limit credit supply. When bank capital is scarce, bank-dependent firms face higher spreads or quantity rationing even though their own fundamentals are unchanged.

(b) Seminal papers

- Holmstrom and Tirole (1997). "Financial Intermediation, Loanable Funds, and the Real Sector." *Quarterly Journal of Economics* 112(3): 663–691.

- Gertler and Kiyotaki (2010). "Financial Intermediation and Credit Policy in Business Cycle Analysis." In *Handbook of Monetary Economics*, Vol. 3.
- Gertler and Karadi (2011). "A Model of Unconventional Monetary Policy." *Journal of Monetary Economics* 58(1): 17–34.
- Adrian and Shin (2010). "Liquidity and Leverage." *Journal of Financial Intermediation* 19(3): 418–437.
- He and Krishnamurthy (2013). "Intermediary Asset Pricing." *American Economic Review* 103(2): 732–770.
- Brunnermeier and Sannikov (2014). "A Macroeconomic Model with a Financial Sector." *American Economic Review* 104(2): 379–421.

(c) Simplified model

Intermediary. Banks have net worth N and can extend total loans L only up to a multiple of net worth:

$$L \leq \kappa N.$$

The constraint may come from regulation, intermediary moral hazard, risk-bearing capacity, or market funding constraints.

Borrowers. Bank-dependent firms demand capital according to:

$$\alpha z_i k_i(s)^{\alpha-1} = r + s,$$

where s is the bank lending spread above the frictionless funding cost.

Aggregate loan capacity. The banking sector must satisfy:

$$\int_i \big(k_i(s) - a_i\big)_+ \, di \leq \kappa N.$$

Let $\psi \geq 0$ be the shadow value of one extra unit of bank lending capacity.

Competitive bank pricing. A marginal loan uses one unit of scarce lending capacity, so its marginal cost to the bank is the frictionless funding cost plus the shadow cost of capacity:

$$r + \psi.$$

With competitive banks and free entry, the lending rate equals marginal cost:

$$r + s = r + \psi.$$

Therefore, in a competitive lending market with scarce bank capital, the spread is:

$$s = \psi, \qquad \lambda_i^{INT} = \psi \quad \text{for bank-dependent borrowers.} \tag{7}$$

This differs from the market-power friction below. Here banks are competitive, so the spread reflects the shadow cost of scarce balance-sheet capacity. In Friction 6, banks have pricing power and choose a markup above marginal cost. When bank net worth N falls, the capacity constraint tightens, ψ rises, and firms face a higher effective cost of capital.

(d) Which firms are constrained

Bank-dependent borrowers: small firms, firms without bond-market access, firms in regions dominated by impaired banks, and firms in concentrated credit markets. The wedge is lender-side: borrower productivity, opacity, or collateral may be unchanged, but the supply of credit contracts.

(e) Policy lever

- Bank recapitalization and stress-tested capital restoration.
- Central bank lending facilities or credit policy that bypasses impaired intermediaries.
- Macroprudential policy and countercyclical capital buffers.
- Regulation of intermediary leverage and maturity mismatch.

(f) What distinguishes this friction

The bottleneck is the lender's balance sheet, not the borrower's pledgeability. Borrower-side reforms such as collateral registries will not solve a pure bank-capital shortage. This is the natural friction for credit crunches and financial-crisis amplification.

Friction 6: Lender Market Power and Relationship Hold-Up

(a) Definition

Lenders with market power can charge markups above the competitive funding cost. In relationship lending, an informed incumbent lender may also extract rents because switching to outside lenders is costly or because outside lenders are less informed.

(b) Seminal papers

- Sharpe (1990). "Asymmetric Information, Bank Lending, and Implicit Contracts: A Stylized Model of Customer Relationships." *Journal of Finance* 45(4): 1069–1087.
- Rajan (1992). "Insiders and Outsiders: The Choice between Informed and Arm's-Length Debt." *Journal of Finance* 47(4): 1367–1400.
- Petersen and Rajan (1995). "The Effect of Credit Market Competition on Lending Relationships." *Quarterly Journal of Economics* 110(2): 407–443.
- Drechsler, Savov, and Schnabl (2017). "The Deposits Channel of Monetary Policy." *Quarterly Journal of Economics* 132(4): 1819–1876.

(c) Simplified model

Monopoly lender. A monopolist lender has cost of funds r and lends to a firm with technology $z_i k_i^\alpha$. At lending rate $\hat{r}$, firm demand is:

$$k_i(\hat{r}) = \left(\frac{\alpha z_i}{\hat{r}}\right)^{\frac{1}{1-\alpha}}.$$

Lender problem.

$$\max_{\hat{r}} \; (\hat{r} - r) k_i(\hat{r}).$$

The elasticity of loan demand is $\varepsilon = 1/(1-\alpha)$. The monopoly markup satisfies:

$$\frac{\hat{r}^* - r}{\hat{r}^*} = \frac{1}{\varepsilon} = 1 - \alpha,$$

so:

$$\hat{r}^* = \frac{r}{\alpha} > r. \tag{8}$$

Input wedge.

$$\lambda_i^{MP} = \hat{r}^* - r = \frac{r(1-\alpha)}{\alpha} > 0. \tag{9}$$

Relationship hold-up. In a relationship-lending version, the wedge is not necessarily a simple monopoly markup. An informed incumbent lender can extract rents because outside lenders cannot distinguish whether the firm is genuinely risky or merely being released by its incumbent bank.

(d) Which firms are constrained

Firms with few alternative lenders, high switching costs, opaque credit histories, or dependence on relationship-specific information. Rural firms and firms in concentrated local banking markets are natural examples.

(e) Policy lever

- Entry of competing lenders and antitrust policy in local credit markets.
- Credit portability, open banking, and registries that reduce informational switching costs.
- Public lending options when private lender concentration is severe.
- Interest-rate caps only with caution: they can reduce markups but may also ration high-risk borrowers.

(f) What distinguishes this friction

The primitive is market structure or relationship hold-up, not asymmetric information alone. Petersen–Rajan (1995) also implies a nuance: more competition need not always improve credit access for young firms if concentrated lenders internalize future relationship rents and therefore lend early. Competition policy can reduce markups, but it may also change dynamic relationship incentives.

Adjacent Module A: Incomplete Contracts and Liquidation

Why this is adjacent rather than core

These models are central to financial contracting, control rights, and bankruptcy. In their cleanest form, the distortion is not a higher marginal cost of initial capital but an inefficient ex post continuation or liquidation decision. The mechanism belongs next to the core taxonomy, but it should not be counted as a seventh direct input-cost wedge unless the model is explicitly recast with an ex ante investment margin.

Seminal papers

- Aghion and Bolton (1992). "An Incomplete Contracts Approach to Financial Contracting." *Review of Economic Studies* 59(3): 473–494.
- Hart and Moore (1998). "Default and Renegotiation: A Dynamic Model of Debt." *Quarterly Journal of Economics* 113(1): 1–41.
- Bolton and Scharfstein (1990). "A Theory of Predation Based on Agency Problems in Financial Contracting." *American Economic Review* 80(1): 93–106.
- Innes (1990). "Limited Liability and Incentive Contracting with Ex-Ante Action Choices." *Journal of Economic Theory* 52(1): 45–67.

Minimal model

Setup. At $t = 0$, the entrepreneur raises debt with face value D. At $t = 1$, a state s is realized. The firm can either continue or be liquidated. Continuation generates total value $V(s)$. Liquidation generates total value L.

Efficient continuation rule. A planner compares total surplus under continuation and liquidation. The efficient rule is:

$$\text{continue if } V(s) > L, \qquad \text{liquidate if } L > V(s).$$

Creditor payoff. Debt gives creditors control in some bad states, for example after default or covenant violation. If creditors liquidate, they receive the liquidation value up to the face value of debt:

$$\text{creditor payoff under liquidation} = \min\{D, L\}.$$

If the firm continues, creditors receive a continuation payoff $C(s)$, which is bounded above by the face value of debt:

$$C(s) \leq D.$$

This continuation payoff may be low in bad states because cash flows are delayed, non-contractible, renegotiated, or hard to pledge.

Creditor liquidation rule. Creditors choose liquidation whenever their private payoff from liquidation exceeds their payoff from continuation:

$$\min\{D, L\} > C(s).$$

This is a private payoff comparison. It is not the same as the efficient comparison between L and $V(s)$.

Inefficient liquidation. Inefficient liquidation occurs in states where creditors prefer liquidation even though continuation creates more total surplus:

$$V(s) > L \qquad \text{but} \qquad \min\{D, L\} > C(s).$$

A simple sufficient case is:

$$V(s) > L \geq D > C(s).$$

In this case, liquidation allows creditors to recover the full face value D, while continuation gives them only $C(s)$. Creditors therefore liquidate, even though total firm value would be higher under continuation.

Expected ex post loss. The expected loss from incomplete contracting is:

$$\Omega^{IC} = \Pr[\text{inefficient liquidation}] \times \mathbb{E}[V(s) - L \mid \text{inefficient liquidation}]\,. \tag{10}$$

Interpretation. The primitive distortion is ex post control rather than an expensive marginal unit of initial capital: creditors may choose liquidation because it maximizes their private recovery, even when continuation maximizes total surplus.

How to recast it as a wedge

If ex ante investment k scales up the continuation surplus and also increases the states in which creditor control destroys value, write expected payoff as:

$$\mathbb{E}[y(k)] - rk - \Omega^{IC}(k).$$

The first-order condition is:

$$\mathbb{E}[y'(k)] = r + \Omega^{IC\prime}(k).$$

Only in this recast version does incomplete contracting generate a marginal investment wedge:

$$\lambda^{IC} = \Omega^{IC\prime}(k).$$

Which firms are affected and policy lever

The affected firms have high continuation value but volatile or hard-to-contract states: restructuring-intensive firms, innovative firms, and firms whose value is mostly going-concern value rather than liquidation value. The policy lever is bankruptcy design, debtor-in-possession finance, renegotiation protocols, and creditor coordination, not simply cheaper credit.

Adjacent Module B: Liquidity and Rollover Risk

Why this is adjacent rather than core

Liquidity and rollover models ask whether firms or intermediaries can finance future liquidity needs after initial investment. The cleanest distortion is inefficient liquidation after a shock or costly precautionary liquidity hoarding. This can be translated into an expected marginal investment wedge, but the primitive is intertemporal insurance rather than the initial cost of funds.

Seminal papers

- Diamond and Dybvig (1983). "Bank Runs, Deposit Insurance, and Liquidity." *Journal of Political Economy* 91(3): 401–419.
- Diamond (1991). "Debt Maturity Structure and Liquidity Risk." *Quarterly Journal of Economics* 106(3): 709–737.
- Holmstrom and Tirole (1998). "Private and Public Supply of Liquidity." *Journal of Political Economy* 106(1): 1–40.

Minimal model

Setup. At $t = 0$, a firm invests k. The investment cannot produce final output unless the firm survives a future liquidity shock. At $t = 1$, the firm must pay a liquidity need ηk, where:

$$\eta \sim F, \qquad \eta \geq 0.$$

If the firm obtains liquidity and continues, it produces final output at $t = 2$:

$$R^c k.$$

If the firm cannot obtain liquidity, it is liquidated early and produces zero continuation value.

Efficient continuation rule. A planner compares the continuation surplus $R^c k$ to the liquidity cost ηk. Since both scale with k, the efficient rule is:

$$\text{continue if } R^c k \geq \eta k,$$

or equivalently:

$$\text{continue if } \eta \leq R^c.$$

Private liquidity constraint. The firm may not be able to raise liquidity for every shock that is socially worth financing. Suppose private liquidity arrangements allow the firm to survive only shocks up to:

$$\eta^* < R^c.$$

Then the firm continues if:

$$\eta \leq \eta^*,$$

and is liquidated if:

$$\eta > \eta^*.$$

Inefficient liquidation. Inefficient liquidation occurs for intermediate liquidity shocks:

$$\eta^* < \eta \leq R^c.$$

In these states, continuation is socially efficient because $R^c > \eta$, but the firm cannot obtain the liquidity needed to continue.

Expected ex post loss. Conditional on a liquidity shock η, the surplus lost from liquidation is:

$$(R^c - \eta)k.$$

Therefore the expected loss from missing liquidity insurance is:

$$\Omega^{LIQ}(k) = k \int_{\eta^*}^{R^c} (R^c - \eta)\, dF(\eta).$$

Define the per-unit expected liquidity loss:

$$\ell(\eta^*) = \int_{\eta^*}^{R^c} (R^c - \eta)\, dF(\eta).$$

Then:

$$\Omega^{LIQ}(k) = k\ell(\eta^*).$$

How to recast it as a wedge

Ex ante, the firm internalizes that some socially valuable projects will be liquidated after liquidity shocks. If expected production is $y(k)$, the ex ante objective can be written as:

$$\mathbb{E}[y(k)] - rk - \Omega^{LIQ}(k).$$

The first-order condition is:

$$\mathbb{E}[y'(k)] = r + \Omega^{LIQ\prime}(k).$$

If the loss is proportional to investment,

$$\Omega^{LIQ}(k) = k\ell(\eta^*),$$

then:

$$\Omega^{LIQ\prime}(k) = \ell(\eta^*),$$

and the investment condition becomes:

$$\mathrm{MRPK}_i = r + \ell(\eta^*).$$

$$\ell(\eta^*) = \int_{\eta^*}^{R^c} (R^c - \eta)\, dF(\eta). \tag{11}$$

So the implied wedge is:

$$\lambda_i^{LIQ} = \ell(\eta^*).$$

Interpretation. The primitive distortion is missing future liquidity insurance rather than a higher interest rate at $t = 0$. A firm may finance the initial investment but still be forced into inefficient liquidation when the future liquidity need is too large to refinance privately. The wedge appears only after recasting the model as an ex ante investment problem.

Which firms are affected and policy lever

Affected firms have long gestation periods, lumpy working-capital needs, large refinancing exposure, or exposure to aggregate liquidity shocks. The policy lever is public liquidity provision, lender-of-last-resort facilities, credit lines, guarantees for interim liquidity needs, and safe public assets that help store liquidity.

Which Papers Use Which Object?

Different papers put the financing friction on different objects. The categories below are best read as modeling choices, since several papers can be rewritten into an equivalent reduced form after substituting the firm's balance sheet identity.

Object in the model and representative papers	Interpretation
Pledgeable productive or collateral asset. $b_i \leq \theta k_i$, or a collateral-value analogue. Hart–Moore (1994), Kiyotaki–Moore (1997), Rampini–Viswanathan (2010), Chaney–Sraer–Thesmar (2012).	Credit capacity comes from assets that creditors can seize or redeploy. Net worth matters because it finances the unpledgeable part of investment.
Direct wealth or net-worth ceiling. $b_i \leq \phi a_i$, or $k_i \leq \bar{k}(a_i)$. Evans–Jovanovic (1989), Banerjee–Newman (1993), Buera–Kaboski–Shin (2011), Moll (2014).	Net worth is the state variable that determines feasible scale. This is often the cleanest reduced form for wealth-constrained entrepreneurship or self-financing dynamics.
Pledgeable-income or incentive ceiling. $\rho k_i \geq r(k_i - a_i)$. Holmstrom–Tirole (1997), Tirole (2006).	Net worth reduces outside finance, while pledgeability comes from incentive-compatible cash flow rather than liquidation value.
Leverage-premium formulation. $\lambda_i = s(k_i/a_i)$. Townsend (1979), Gale–Hellwig (1985), Bernanke–Gertler (1989), Bernanke–Gertler–Gilchrist (1999).	Net worth reduces leverage and therefore lowers the external finance premium. The friction is a smooth price wedge rather than a hard investment ceiling.

The analytical trade-off is clearest in dynamic models. A direct wealth ceiling makes a_i the relevant state and gives a simple feasible set for capital choice. A pledgeable-capital constraint is more structural, but it can require solving for collateral values, asset prices, or continuation values jointly with investment. This extra structure is valuable when the question is about collateral prices or creditor rights, but it is not always needed when the goal is to study how limited net worth slows convergence to efficient scale.

Decision Guide

If your model needs...	Use	Because...
A tractable wealth-based borrowing ceiling	1	Limited enforcement gives the cleanest $k \leq \bar{k}(a)$ constraint
To microfound pledgeability from hidden action	2	$B/\Delta p$ determines how much cash flow can be pledged
An external finance premium increasing in leverage	3	Costly verification gives a smooth premium $s(k/a)$
Credit denial rather than just expensive credit	4	Adverse selection makes lender returns non-monotone in the rate
A credit crunch from bank losses	5	The lender's balance sheet, not the borrower, is the bottleneck
Bank competition, local market concentration, or hold-up	6	The wedge is a markup or switching-cost rent
Bankruptcy or inefficient liquidation	A	The primary distortion is ex post continuation control
Government liquidity provision or rollover risk	B	The primary distortion is missing future liquidity insurance

Summary: Distinguishing the Core Frictions

#	Friction	Which firms?	Policy lever	Key parameter
1	Limited enforcement	Low net worth; hard-to-pledge assets; low-wealth entrepreneurs	Creditor rights, registries, titling, guarantees	θ
2	Moral hazard	Opaque cash flows; weak governance; low net worth	Monitoring, governance, incentive design	$B/\Delta p$
3	Costly state verification	High leverage; hard-to-audit output	Accounting, disclosure, audit technology	γ, k/a
4	Adverse selection	Observationally opaque applicants; no track record	Credit registries, screening, careful guarantees	Pool composition $\mu(\hat{r})$
5	Intermediary constraints	Bank-dependent borrowers	Recapitalization, credit facilities, macroprudential policy	Bank net worth N
6	Lender market power	Few alternative lenders; high switching costs	Entry, portability, competition policy	Markup / outside option

Module	Mechanism	Primary distortion	Policy lever
A	Incomplete-contract liquidation	Inefficient ex post continuation or liquidation	Bankruptcy design, renegotiation, creditor coordination
B	Liquidity / rollover risk	Missing future liquidity insurance and inefficient liquidation	Public liquidity, credit lines, lender of last resort

References

- Adrian and Shin (2010). "Liquidity and Leverage." *Journal of Financial Intermediation* 19(3): 418–437.
- Aghion and Bolton (1992). "An Incomplete Contracts Approach to Financial Contracting." *Review of Economic Studies* 59(3): 473–494.
- Akerlof (1970). "The Market for 'Lemons'." *Quarterly Journal of Economics* 84(3): 488–500.
- Albuquerque and Hopenhayn (2004). "Optimal Lending Contracts and Firm Dynamics." *Review of Economic Studies* 71(2): 285–315.
- Banerjee and Newman (1993). "Occupational Choice and the Process of Development." *Journal of Political Economy* 101(2): 274–298.
- Bernanke and Gertler (1989). "Agency Costs, Net Worth, and Business Fluctuations." *American Economic Review* 79(1): 14–31.
- Bernanke, Gertler, and Gilchrist (1999). "The Financial Accelerator in a Quantitative Business Cycle Framework." In *Handbook of Macroeconomics*, Vol. 1.
- Bester (1985). "Screening vs. Rationing in Credit Markets with Imperfect Information." *American Economic Review* 75(4): 850–855.
- Bolton and Scharfstein (1990). "A Theory of Predation Based on Agency Problems in Financial Contracting." *American Economic Review* 80(1): 93–106.
- Brunnermeier and Sannikov (2014). "A Macroeconomic Model with a Financial Sector." *American Economic Review* 104(2): 379–421.
- Buera, Kaboski, and Shin (2011). "Finance and Development: A Tale of Two Sectors." *American Economic Review* 101(5): 1964–2002.
- Chaney, Sraer, and Thesmar (2012). "The Collateral Channel: How Real Estate Shocks Affect Corporate Investment." *American Economic Review* 102(6): 2381–2409.
- de Meza and Webb (1987). "Too Much Investment: A Problem of Asymmetric Information." *Quarterly Journal of Economics* 102(2): 281–292.
- Diamond (1991). "Debt Maturity Structure and Liquidity Risk." *Quarterly Journal of Economics* 106(3): 709–737.
- Diamond and Dybvig (1983). "Bank Runs, Deposit Insurance, and Liquidity." *Journal of Political Economy* 91(3): 401–419.
- Drechsler, Savov, and Schnabl (2017). "The Deposits Channel of Monetary Policy." *Quarterly Journal of Economics* 132(4): 1819–1876.
- Evans and Jovanovic (1989). "An Estimated Model of Entrepreneurial Choice under Liquidity Constraints." *Journal of Political Economy* 97(4): 808–827.
- Gale and Hellwig (1985). "Incentive-Compatible Debt Contracts: The One-Period Problem." *Review of Economic Studies* 52(4): 647–663.
- Gertler and Karadi (2011). "A Model of Unconventional Monetary Policy." *Journal of Monetary Economics* 58(1): 17–34.
- Gertler and Kiyotaki (2010). "Financial Intermediation and Credit Policy in Business Cycle Analysis." In *Handbook of Monetary Economics*, Vol. 3.
- Hart and Moore (1994). "A Theory of Debt Based on the Inalienability of Human Capital." *Quarterly Journal of Economics* 109(4): 841–879.
- Hart and Moore (1998). "Default and Renegotiation: A Dynamic Model of Debt." *Quar-*

terly Journal of Economics 113(1): 1–41.
- He and Krishnamurthy (2013). "Intermediary Asset Pricing." *American Economic Review* 103(2): 732–770.
- Holmstrom and Tirole (1997). "Financial Intermediation, Loanable Funds, and the Real Sector." *Quarterly Journal of Economics* 112(3): 663–691.
- Holmstrom and Tirole (1998). "Private and Public Supply of Liquidity." *Journal of Political Economy* 106(1): 1–40.
- Hsieh and Klenow (2009). "Misallocation and Manufacturing TFP in China and India." *Quarterly Journal of Economics* 124(4): 1403–1448.
- Innes (1990). "Limited Liability and Incentive Contracting with Ex-Ante Action Choices." *Journal of Economic Theory* 52(1): 45–67.
- Jaffee and Russell (1976). "Imperfect Information, Uncertainty, and Credit Rationing." *Quarterly Journal of Economics* 90(4): 651–666.
- Jensen and Meckling (1976). "Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure." *Journal of Financial Economics* 3(4): 305–360.
- Kehoe and Levine (1993). "Debt-Constrained Asset Markets." *Review of Economic Studies* 60(4): 865–888.
- Kiyotaki and Moore (1997). "Credit Cycles." *Journal of Political Economy* 105(2): 211–248.
- Modigliani and Miller (1958). "The Cost of Capital, Corporation Finance, and the Theory of Investment." *American Economic Review* 48(3): 261–297.
- Moll (2014). "Productivity Losses from Financial Frictions: Can Self-Financing Undo Capital Misallocation?" *American Economic Review* 104(10): 3186–3221.
- Myers and Majluf (1984). "Corporate Financing and Investment Decisions When Firms Have Information That Investors Do Not Have." *Journal of Financial Economics* 13(2): 187–221.
- Petersen and Rajan (1995). "The Effect of Credit Market Competition on Lending Relationships." *Quarterly Journal of Economics* 110(2): 407–443.
- Rajan (1992). "Insiders and Outsiders: The Choice between Informed and Arm's-Length Debt." *Journal of Finance* 47(4): 1367–1400.
- Rampini and Viswanathan (2010). "Collateral, Risk Management, and the Distribution of Debt Capacity." *Journal of Finance* 65(6): 2293–2322.
- Sharpe (1990). "Asymmetric Information, Bank Lending, and Implicit Contracts: A Stylized Model of Customer Relationships." *Journal of Finance* 45(4): 1069–1087.
- Stiglitz and Weiss (1981). "Credit Rationing in Markets with Imperfect Information." *American Economic Review* 71(3): 393–410.
- Tirole (2006). *The Theory of Corporate Finance.* Princeton University Press.
- Townsend (1979). "Optimal Contracts and Competitive Markets with Costly State Verification." *Journal of Economic Theory* 21(2): 265–293.